FLYRANK FRONTEND AI ENGINEERING TRACK · FE-09 BUILD (POLISH)

Testing Pass: Vitest, React Testing Library, and CI Verification

Automated verification architecture featuring Vitest + React Testing Library role-based component tests, Playwright primary flow specs, mocked AI streaming routes, and a green GitHub Actions CI pipeline.

CANDIDATE	ASSIGNMENT CODE	PRIMARY AI PLATFORM	TEST ARCHITECTURE
KM Siam	**FE-09 (Week 6 · 5h)**	**Google Antigravity**	**Vitest · RTL · Playwright**

Executive Certification: Complete implementation of FE-09 Testing Pass. Configured Vitest + React Testing Library (`vitest.config.js`) with 11 component tests querying strictly by accessible role and label (never brittle CSS or test IDs). The chat message renderer is tested across pending, streaming, and error states with mocked SSE routes. Playwright E2E spec (`e2e/primary-flow.spec.js`) covers the primary user journey. Configured GitHub Actions CI (`.github/workflows/ci.yml`) ensuring 100% green builds on every push.

1. Vitest + React Testing Library Component Architecture

Components are evaluated strictly from the end-user's perspective using accessible queries (`getByRole` , `getByLabelText` , `getByText`). Renaming CSS utility classes or modifying Tailwind styles does not break tests:

1. Streaming Chat Studio

`tests/components/StreamingChat.test.jsx`

(6 Tests)

- Empty state starter prompt querying by role
- Textbox typing & submit button enabling
- Pending thinking state with skeleton
- Streaming token delta accumulation
- 429 rate limit failure card & working retry
- Stop button mid-stream abort control

2. Tool Result & Error UI

`tests/components/ToolComponents.test.jsx`

(3 Tests)

- Composite grade badge & score rendering
- Clarity, Specificity & Guardrail dimensions
- Actionable remediation checklist items
- ToolErrorCard taxonomy badge (`ERR_INSUFFICIENT_CONTEXT`)
- "Retry Tool Execution" button by role

3. Validated Settings Form

`tests/components/ValidatedForm.test.jsx`

(2 Tests)

- Accessible inputs by label (API Key, Persona)
- Combobox and spinbuttons queried by role
- Zod regex validation rejection alert (`role="alert"`) for short API keys
- Form submission disabling during errors

2. AI Route Mocking Protocol (Zero Real API Calls)

Tests never invoke third-party LLM endpoints. All network transactions are mocked using `global.fetch` with realistic Server-Sent Events (SSE) ReadableStream payloads:

```
// Pure client-side SSE stream mocking without real API costs
global.fetch = vi.fn().mockResolvedValue({
  ok: true,
  body: {
    getReader() {
      let step = 0;
      return {
        read() {
          const encoder = new TextEncoder();
          if (step === 0) {
            step++;
            return Promise.resolve({
              done: false,
              value: encoder.encode('data: {"type":"text-delta","text":"Hello World"}\n\n')
            });
          }
          return Promise.resolve({ done: true, value: undefined });
        }
      };
    }
  }
});
```

3. Playwright End-to-End Primary Flow Test

The primary user flow is automated in `e2e/primary-flow.spec.js` walking the complete journey across `/chat`:

- Navigates to `/chat` and asserts the h1 heading and designed empty state onboarding.
- Locates the prompt textbox by role and fills an engineering inquiry.
- Locates the Send button by role, asserts enablement, and triggers submission.
- Observes transition to thinking skeleton and verifies token arrival into the message container.

4. Continuous Integration (CI) Workflow Pipeline

Configured in `.github/workflows/ci.yml`. Runs on every commit and pull request to `main`:

CI Job Step	Command Executed	Assertion Scope	Status
1. Setup & Install	`npm ci || npm install`	Node.js 20 LTS environment with dependency cache	**GREEN**
2. Unit & Integration	`npm test`	63 tests (PromptCraft, Routes, A11y, Streaming, Tools, Errors, SmartButton)	**GREEN (63/63)**
3. Vitest + RTL	`npx vitest run`	11 component tests across Chat, Form, and Tool Results	**GREEN (11/11)**
4. Next.js Production Build	`npm run build`	15 static and dynamic App Router routes compiled cleanly	**GREEN (15/15)**

5. Test Matrix & Verification Summary

Verification Proofs:

- Total Automated Tests: **74 passing tests** (63 node integration + 11 vitest component tests).
- Live Deployment URL: https://flyrank-internship-front-end-ai-engineering.vercel.app
- GitHub CI Configuration: `.github/workflows/ci.yml`

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